Sam Belliveau

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Education

Cornell University — Ithaca, NY
B.S. in Electrical & Computer Engineering — *Expected May 2026*

- **GPA:** 3.7 / 4.3

Languages: C, C++, C#, Python, Java, Rust, Swift, Go, JavaScript, Verilog HDL, SQL, HTML, GLSL, Bash / ZSH
Technologies: Git, Linux, Docker, VS Code, Robot OS (ROS), Cassandra, BigQuery, Numpy, Scipy, PyTorch, GenAI

Experience

Robotics Software Engineer — November 2023 – Present
Cornell University Autonomous Underwater Vehicles — *Ithaca, NY*

- Slashed CPU usage by 80% by offloading Kalman Filter computations to the GPU using Nvidia's CUDA framework
- Enhanced autonomous navigation precision by implementing least squares optimization for dynamic thruster allocation

Signal Analysis Intern — January 2023 – May 2023
Feinstein Institute for Medical Research — *Manhasset, NY*

- Engineered an automated Sharp Wave Ripple (SWR) detection pipeline using Python and SciPy to process large-scale EEG datasets
- Conducted analysis on SWR characteristics to refine detection algorithms, directly contributing to improved seizure prediction models

Software Intern on Consumer Product Team — July 2022 – September 2022
Reddit Inc. — *Remote*

- Collaborated with the Taxonomy Group to classify and ensure the safety of over 138,000 subreddits
- Designed and implemented a synthetic data generation script to rigorously validate the accuracy of safety classification models
- Optimized content moderation workflows by integrating real-time statistical metrics into the dashboard, accelerating the identification of problematic content

Software Intern on Consumer Product Team — July 2021 – September 2021
Reddit Inc. — *Remote*

- Authored comprehensive unit tests to ensure code reliability and prevent regressions prior to production deployment
- Leveraged BigQuery and Cassandra to architect a scalable notification system for moderator engagement campaigns

Projects

UCI Chess Engine in Rust | *Rust, Min Max, Alpha Beta Pruning, Chess* — June 2023 – July 2023

- Developed a high-performance UCI-compliant Chess Engine in Rust, achieving an estimated Elo of 2700 (Grandmaster level)
- Optimized search efficiency using Alpha-Beta Pruning and iterative deepening, significantly boosting engine strength

Dolphin Emulator | *C++, GLSL, JIT Recompilation* — September 2022 – Present

- Optimized emulation performance on low-end hardware by dynamically scaling video interrupt signal frequency
- Implemented six GPU-accelerated resolution scaling algorithms (GLSL), enhancing image fidelity across diverse display resolutions